COMP4451 Project 3 - Weights & Biases Experiment Report

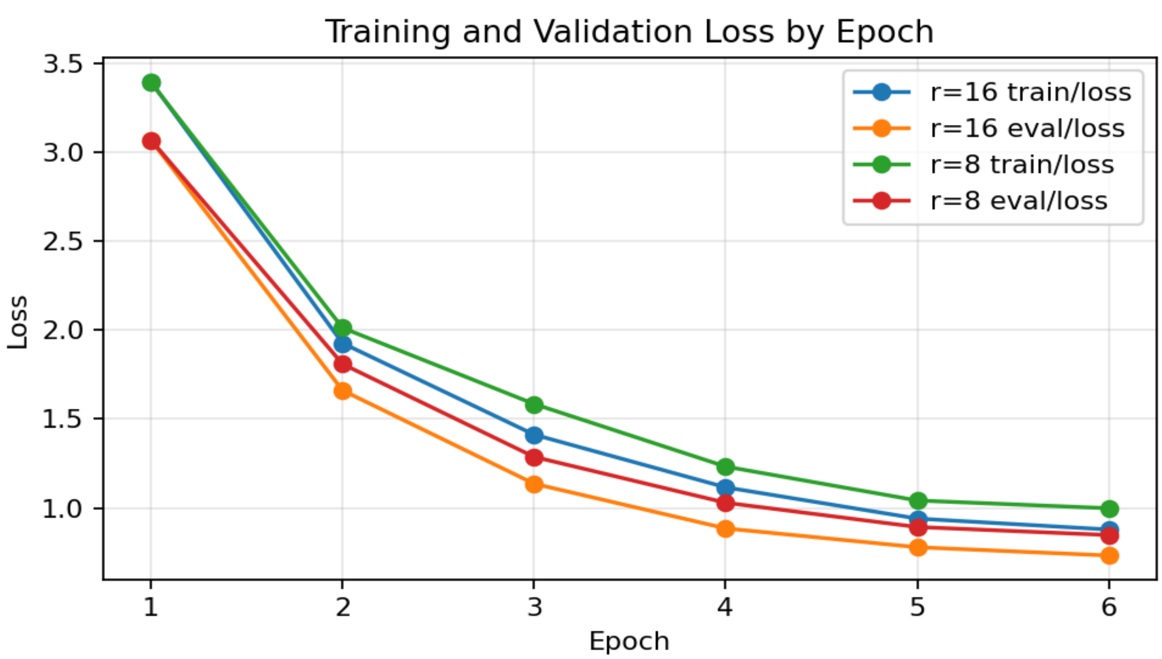
BANKING77 ELECTRA + LoRA Intent Classification

Generated from the completed W&B; run outputs and dashboard URLs supplied in the notebook. This report includes the required loss curve, validation F1 curve, learning-rate schedule, and run summary comparison table.

Metric	Main r=16	Comparison r=8
Run name	electra-lora_r16_lr0.0001_ep6	electra-lora_r8_alpha32_lr0.0001_ep6
Dashboard URL	W&B run 42xe8vam	W&B run g4ho1qga
LoRA rank	16	8
LoRA scaling alpha/r	2.00	4.00
Trainable parameters	1,534,541 (1.382%)	1,092,173 (0.987%)
Validation F1	0.7979	0.7831
Test F1	0.7934	0.7662
Validation-test gap	0.0079	0.0179

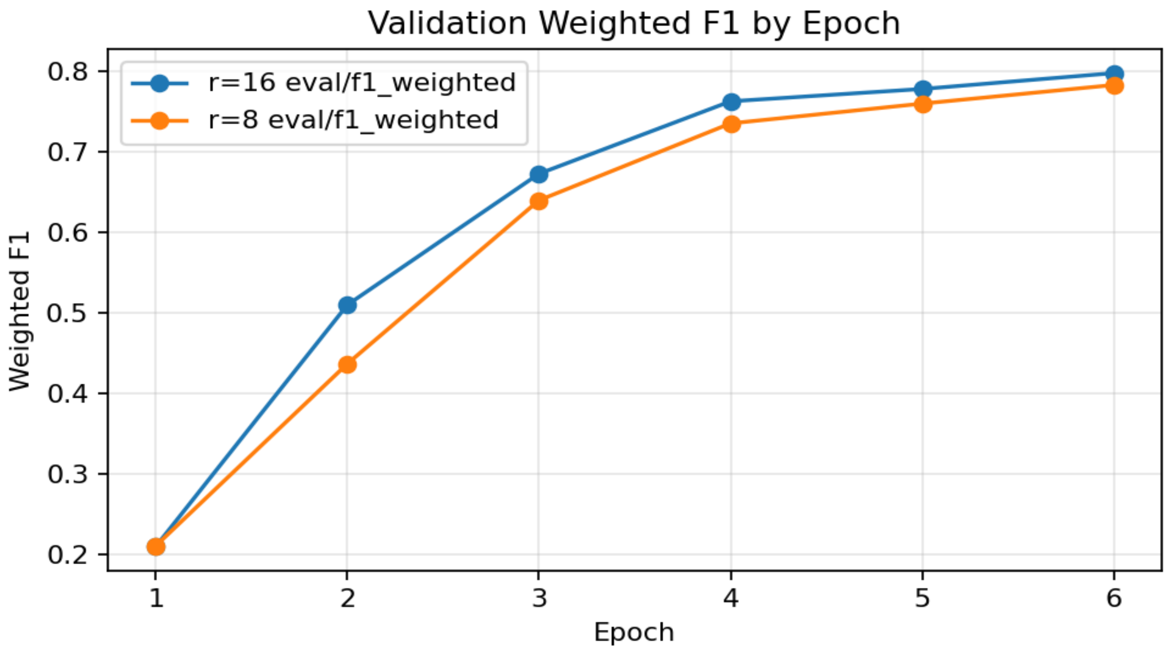
Interpretation: The r=16 adapter is the stronger final configuration. It achieved higher validation and test F1 while still updating only 1.382% of the model parameters. The r=8 run is more parameter-efficient but gave up 0.0148 validation F1 and 0.0272 test F1.

Training and Validation Loss



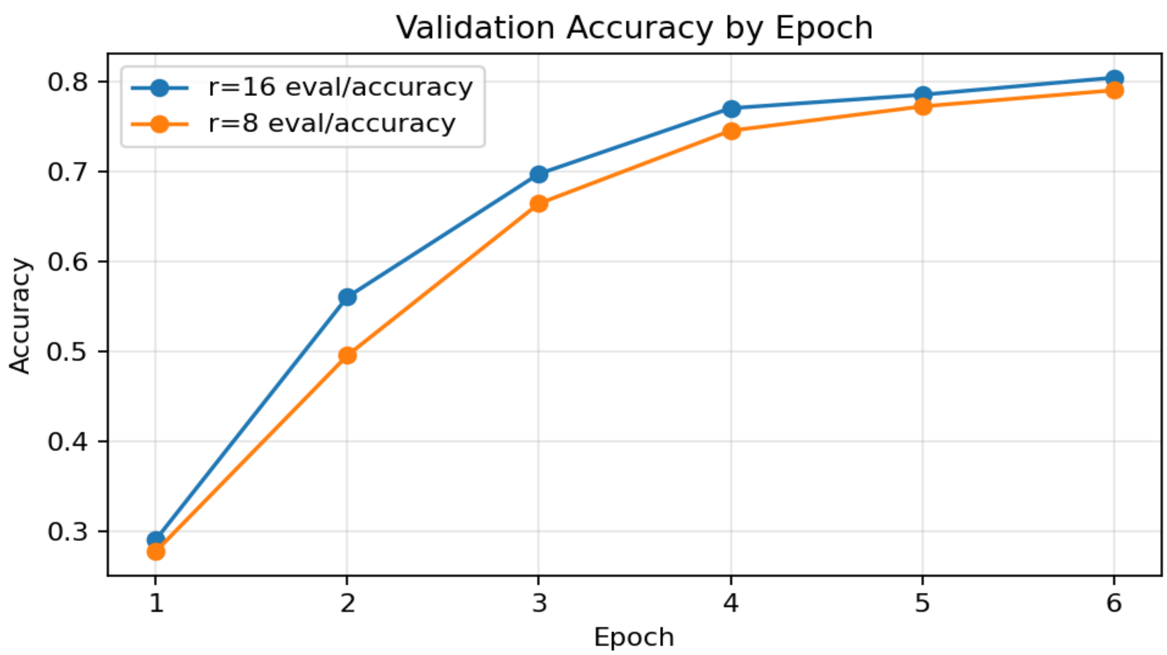
Both runs show healthy decreasing loss. The r=16 run maintains lower validation loss by the final epoch.

Validation Weighted F1



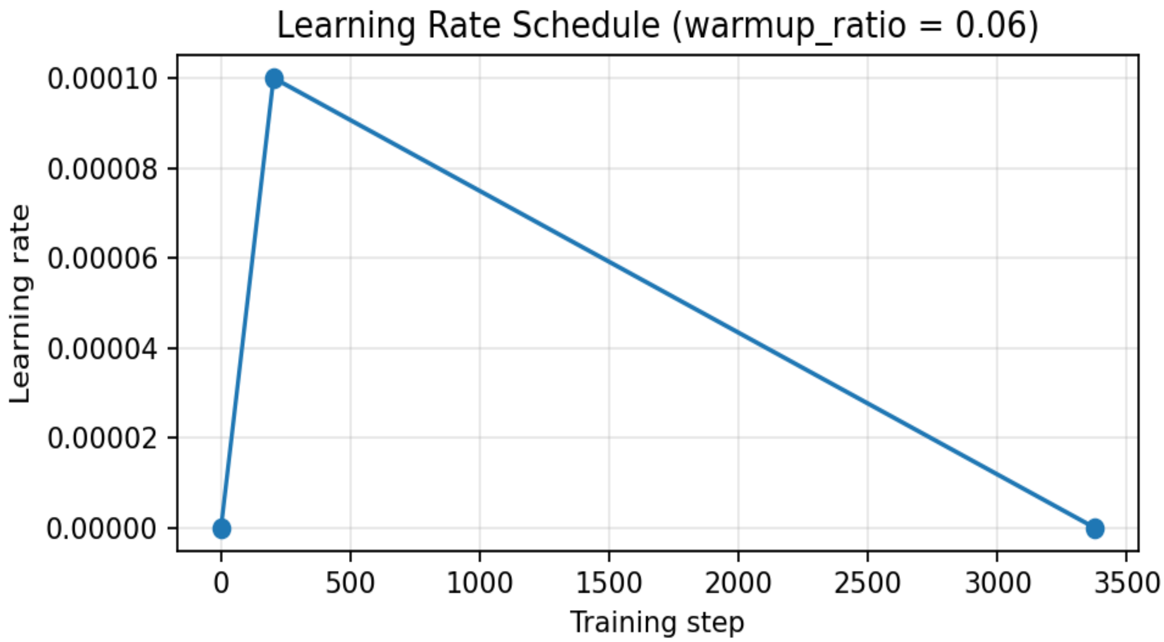
F1 rises every epoch for both runs. The r=16 model finishes at 0.7979, outperforming the r=8 model at 0.7831.

Validation Accuracy



Validation accuracy improves monotonically across epochs. The main r=16 run reaches 0.8050.

Learning Rate Schedule



With warmup_ratio = 0.06, the schedule ramps up early and then decays linearly, reducing early training instability.

Epoch-Level Metric Tables

Main run: r=16, alpha=32

Epoch	Train Loss	Val Loss	Val Acc	Val F1
1	3.3970	3.0681	0.2910	0.2098
2	1.9261	1.6610	0.5610	0.5100
3	1.4120	1.1359	0.6980	0.6728
4	1.1155	0.8847	0.7710	0.7628
5	0.9400	0.7788	0.7860	0.7782
6	0.8783	0.7323	0.8050	0.7979

Comparison run: r=8, alpha=32

Epoch	Train Loss	Val Loss	Val Acc	Val F1
1	3.3918	3.0689	0.2780	0.2107
2	2.0124	1.8091	0.4960	0.4368
3	1.5844	1.2869	0.6650	0.6399
4	1.2324	1.0291	0.7460	0.7354
5	1.0417	0.8924	0.7730	0.7600
6	0.9979	0.8473	0.7910	0.7831

Final Selection

I select the r=16 run as the final model. It has the best validation F1, best test F1, smaller validation-test gap, and still satisfies the PEFT goal by updating a small fraction of model parameters. The r=8 run demonstrates a useful efficiency tradeoff, but it underperforms the r=16 run on the metrics most relevant for assignment evaluation.